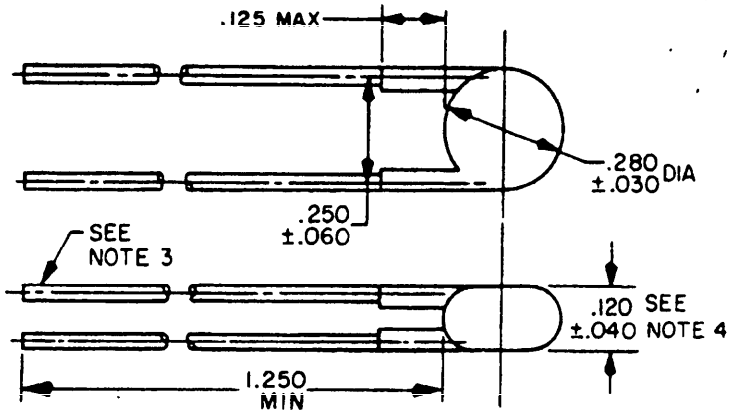
MILITARY SPECIFICATION SHEET

CAPACITORS, FIXED, CERAMIC DIELECTRIC (TEMPERATURE COMPENSATING), ESTABLISHED AND NON-ESTABLISHED RELIABILITY, STYLES CCR54 AND CC54

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the capacitors described herein shall consist of this document and the latest issue of Specification MIL-C-20.



INCHES	MM
.002	.05
.025	.64
.030	.76
.040	1.02
.060	1.52
.120	3.05
.125	3.18
.187	4.75
.250	6.35
.280	7.11
1.250	31.75

NOTES:
1. Dimensions are in inches.
2. Metric equivalents (to the nearest .01 mm) are given for general information only and are based upon 1 inch = 25.4 mm.
3. Lead diameter shall be .025 ± .002.
4. For capacities 1 to 3 pF dimension shall be .187 ± .030.

FIGURE 1. Styles CCR54 and CC54 capacitors.

REQUIREMENTS:

Design and construction:
Dimensions and configuration: See figure 1.
Lead type: Radial.
Case type: Disk.
DC rated voltage: 500 volts.
Operating temperature range: -55° to +125° C.
Characteristic: See table I.
Failure rate level (CCR54 only): M (1.0%) or P (0.1%).

Temperature cycling and voltage conditioning (CCR54 only): In accordance with MIL-C-20.

Capacitance: Within tolerance specified (see table I).

Dissipation factor: In accordance with MIL-C-20.

Dielectric withstanding voltage: In accordance with MIL-C-20.
Body insulation - Test II.

FSC 5910

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TABLE I. Capacitor characteristics.

Type designation 1/	Rated voltage	Nom. cap.	Cap. tol.	Type designation 1/	Rated voltage	Nom. cap.	Cap. tol.
	(volts, dc)	(pF)			(volts, dc)	(pF)	
CC-54CH1R0--	500	1.0	CD	CC-54CH4R7--	500	4.7	CD
CC-54CH1R2--	500	1.2	CD	CC-54UJ5R0--	500	5.0	C
CC-54CH1R5--	500	1.5	CD	CC-54CH5R6--	500	5.6	CD
CC-54CH1R8--	500	1.8	CD	CC-54CH6R8--	500	6.8	CD
CC-54CK2R0--	500	2.0	B	CC-54CH8R2--	500	8.2	CD
CC-54UK2R0--	500	2.0	C	CC-54CH100--	500	10	CD
CC-54CH2R2--	500	2.2	CD	CC-54CG120--	500	12	GJK
CC-54CH2R7--	500	2.7	CD	CC-54HG120--	500	12	F
CC-54CH3R3--	500	3.3	CD	CC-54CG150--	500	15	GJK
CC-54CH3R9--	500	3.9	CD	CC-54HG180--	500	18	F

1/ Complete type designation will include the following:
1st dash - Symbol "R" (for style CCR54) or dash will be deleted (for style CC54).
2nd dash - Applicable capacitance tolerance symbol.
3rd dash - Applicable failure rate level symbol (CCR54 only) or dash will be deleted (for style CC54).

Temperature coefficient and capacitance drift: In accordance with MIL-C-20. The temperature-coefficient tolerance shall be determined in accordance with the curve shown on figure 2. For points below the curve, characteristics are not considered practicably measurable.

Solderability: In accordance with MIL-C-20.

Resistance to soldering heat: In accordance with MIL-C-20.

Life: In accordance with MIL-C-20, operating condition 2.

Marking: Method I of MIL-STD-1285, as shown in following example:

JCH - "J" brand (J); and characteristic (CH).
2R2 - Capacitance value.
CM3 - Cap. tolerance (C); FR level, CCR54 only (M); and year code (3 for 1973).
ATM - Lot code (A); and trademark (TM).

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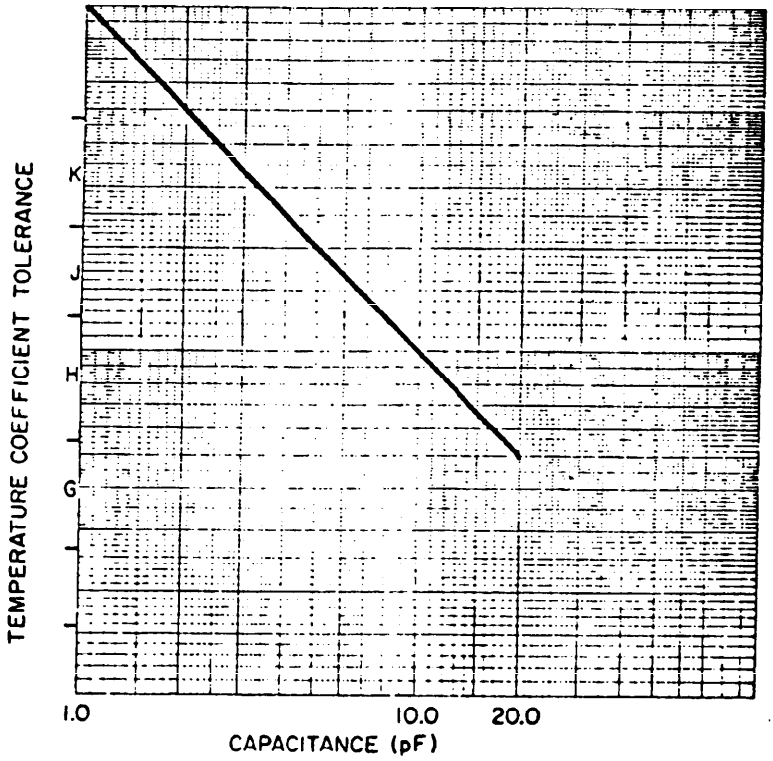


FIGURE 2. Limits for characteristic measurement.

NOTE: Recommended replacement capacitors for those covered in MIL-C-0020/16C(USAF) are as follows:

Recommended type designation 1/ (MIL-C-20/43)	Canceled type designation (MIL-C-0020/16C(USAF))
CC-54CK2R0B	CC50CK2R0B
CC-54UK2R0C	CC50UK2R0C
CC-54UJ5R0C	CC50UK5R0C
CC-54CG120F	CC50HG120F
CC-54HG180F	CC50HG180F

1/ "CCR" (ER part) or "CC" (non-ER part).

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Custodians:
Army - EL
Navy - EC
Air Force - 11

Review activities:
Army -
Navy - OS
Air Force - 80
DSA - ES

User activities:
Army -
Navy - AS, SH, MC
Air Force -

Preparing activity:
Air Force - 11

Agent:
DSA - ES

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